Uncertain Topology

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Abstract

This paper extends Scenario-Based Uncertain Set Theory into the domain of general topology. By avoiding unnecessary symbolic inflation and returning to First Principles, we define an Uncertain Topology simply as a topological structure constructed directly on an Uncertain Universal Space X. We show that every open set inherits state-dependent superposition through its Configuration Space $\mathrm{Conf}(A)$. Crucially, we establish the Topological Collapse Theorem: when the uncertain cardinalities of constituent open sets collapse into deterministic integers ($\mathrm{ucard}(A) \in \mathbb{N}$), the uncertain topology degenerates canonical-wise into a classical static topology. From this collapse mechanism, we propose a lightweight, intuitive metric—the Topological Uncertainty Index $\mu_U(\mathcal{T})$—which measures the precise ratio of deterministic capacity over scenario-expanded uncertainty capacity without requiring complex probabilistic densities or fuzzy membership functions.

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Chapter 1

Uncertain Sets Review

Definition 1.1 (Uncertain Element)**.** *Let U be a non-empty universal set. An uncertain element x on U is an object whose exact value in U is undetermined, constrained only by a non-empty set of possible values $A \subseteq U$:*

$$x \in A$$

The set A is called the possibility domain (or state space) of x, denoted by $\mathrm{Val}(x) = A$.

Definition 1.2 (Uncertain Set)**.** *Let U be a non-empty universal set and I be an index set. An uncertain set S on U is an indexed family of uncertain elements on U:*

$$S = \{x_i : i \in I\}$$

where each uncertain element x_i ($i \in I$) has a possibility domain $\mathrm{Val}(x_i) \subseteq U$.

Definition 1.3 (Configuration Space)**.** *Let $S = \{x_i : i \in I\}$ be an uncertain set on U. A realized configuration (or empirical state) of S is a classical set $S^* \subseteq U$ obtained when all constituent elements x_i simultaneously realize specific values $v_i \in \mathrm{Val}(x_i)$:*

$$S^* = \{v_i : i \in I\}$$

The collection of all classical sets realized by S is called the configuration space of S, denoted by $\mathrm{Conf}(S)$.

Definition 1.4 (Uncertain Cardinality)**.** *Let $S = \{x_i : i \in I\}$ be an uncertain set on U. The uncertain cardinality of S, denoted by* $\mathrm{ucard}(S)$*, is an uncertain number on $\mathbb{N}$ whose possibility domain consists of the classical cardinalities of all realized states in its configuration space* $\mathrm{Conf}(S)$*:*

$$\mathrm{Val}(\mathrm{ucard}(S)) = \{\mathrm{card}(S^*) : S^* \in \mathrm{Conf}(S)\} \subseteq \mathbb{N}$$

Chapter 2

Uncertain Topology

Definition 2.1 (Uncertain Topology)**.** *Let X be an uncertain set serving as the universal space. An* Uncertain Topology *on X, denoted by $\mathcal{T}$, is a collection of uncertain subsets of X satisfying the standard classical topology axioms across all scenario realizations:*

i) $\emptyset \in \mathcal{T}$ *and* $X \in \mathcal{T}$.

ii) The arbitrary union of open uncertain sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

iii) The finite intersection of open uncertain sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

Each element $A \in \mathcal{T}$ is called an uncertain open set*. Its physical state in any specific scenario is completely characterized by its configuration space* $\mathrm{Conf}(A)$.

Theorem 2.1 (Topological Collapse Theorem)**.** *Let $\mathcal{T}$ be an uncertain topology on an uncertain universe X. If for every uncertain open set $A \in \mathcal{T}$, its uncertain cardinality degenerates to a single deterministic natural number:*

$$\mathrm{Val}(\mathrm{ucard}(A)) = \{k_A\}, \quad k_A \in \mathbb{N}$$

then the uncertainty across all scenarios vanishes, and $\mathcal{T}$ degenerates canonically into a classical static topology on a constant universe.

Proof. By definition, $\mathrm{Val}(\mathrm{ucard}(A)) = \{\mathrm{card}(A^*) : A^* \in \mathrm{Conf}(A)\}$. If $\mathrm{Val}(\mathrm{ucard}(A)) = \{k_A\}$, then every realized configuration $A^* \in \mathrm{Conf}(A)$

has identical classical cardinality k_A. This implies that element realization does not induce structural state-collapse or cardinality fluctuation across scenarios. Consequently, $\mathrm{Conf}(A)$ reduces to a singleton configuration set $\{A^*\}$. Thus, every $A \in \mathcal{T}$ behaves identically as a classical set, collapsing $\mathcal{T}$ directly into a classical topology. □

Chapter 3

Degree of Topological Uncertainty

Definition 3.1 (Topological Uncertainty Index)**.** *Let $\mathcal{T}$ be an uncertain topology on X, and let* $\mathrm{Conf}(\mathcal{T})$ *be the ensemble of realized classical topologies induced across scenario space. Let $\mathcal{T}^*_{\mathrm{det}}$ be a canonical deterministic baseline configuration of the topology.*

The Topological Uncertainty Index $\mu_U(\mathcal{T}) \in [0, 1)$ *is defined as the ratio of deterministic cardinality capacity over total uncertain scenario-expanded cardinality capacity:*

$$\mu_U(\mathcal{T}) = 1 - \frac{\sum_{A^* \in \mathcal{T}^*_{\mathrm{det}}} \mathrm{card}(A^*)}{\sum_{A \in \mathcal{T}} \sum_{A^* \in \mathrm{Conf}(A)} \mathrm{card}(A^*)}$$

Proposition 3.1 (Properties of $\mu_U(\mathcal{T})$)**.** *The metric $\mu_U(\mathcal{T})$ satisfies:*

a) $\mu_U(\mathcal{T}) = 0$ *if and only if $\mathcal{T}$ is a classical static topology.*

b) $\mu_U(\mathcal{T}) \to 1$ *as the state-space cardinality fluctuation expands infinitely across scenarios.*

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